

Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented aerospace reliability problem: BGA solder joints in satellite electronics accumulate thermo-mechanical fatigue damage from repeated thermal cycling between sunlight and eclipse (~-150C to +120C per orbit, ~5500 cycles/year in LEO), eventually cracking and producing intermittent or permanent open circuits. The failure is field-documented across aerospace electronics heritage programs and is a dominant driver of avionics derating and redundancy mass. This description is qualitative problem-domain context from the engineering literature; no specific incident statistics are asserted here (the engine's own retrieval must find and custody the evidence).

Mechanism: measures to improve the reliability of solder joints

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Implement lead-free solder joint reinforcement techniques that enhance thermo-mechanical fatigue resistance without adding significant mass or compromising heat dissipation

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: CONDITIONAL; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: NOT ESTABLISHED.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.